# ASHUTOSH BARI

Software Developer | AI/ML Engineer | Agentic AI & LLM Systems

Pune, Maharashtra, India | +91-9921589619 | ashutoshbariofficial@gmail.com

github.com/ashutoshbari | linkedin.com/in/ashutoshbari | ashutoshbari.github.io/Portfolio

## PROFESSIONAL SUMMARY

AI/ML Engineer and Software Developer with 1+ year of full-time, production experience building real-time computer vision systems (YOLOv8, OpenCV, Python) and LLM-powered applications for government and enterprise use cases. Hands-on experience integrating LLMs (Gemini API) into multi-step, stateful conversational workflows, including a 7-stage adaptive interview agent that dynamically generates questions and evaluates responses in real time. Strong Python fundamentals, REST API design (Flask, FastAPI), and experience across the full ML lifecycle — from dataset preparation and model training to deployment. Comfortable working with AI-assisted development tools and quickly picking up new frameworks to ship production-quality systems.

## TECHNICAL SKILLS

- Programming & ML: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- Computer Vision & AI: YOLOv8, OpenCV, Object Detection, Vehicle Tracking, OCR (PaddleOCR), Camera Calibration
- LLM & Agentic AI: LLM Integration (Gemini API), Prompt Engineering, Multi-Step Conversational Agent Design, REST API & Tool Integration for Agents
- Full-Stack & MLOps: Flask, FastAPI, Streamlit, Next.js, REST APIs, Docker, Linux (CLI), Git/GitHub
- Tools & Data: Cursor, Claude AI, ChatGPT, Antigravity, MySQL, SQL, Power BI, Tableau

## PROFESSIONAL EXPERIENCE

**Futops Technologies India Pvt. Ltd.** *1 Year (Full-Time)*

- Built real-time computer vision systems using YOLOv8, OpenCV, and Python for live government surveillance and traffic-monitoring projects.
- Handled data collection and dataset preparation, and trained/tuned custom object detection models — improved model accuracy from ~60% to ~95% through iterative training and hyperparameter tuning.
- Built a vehicle counting and classification system to detect and categorize vehicle types (car, bike, truck, bus) from live traffic camera feeds.
- Developed an ANPR (Automatic Number Plate Recognition) system using PaddleOCR to detect and extract vehicle number plates, used to generate traffic violation challans.
- Coordinated multiple independent detection modules — wrong-direction driving, triple-seat riding, speed detection via camera calibration, and no-parking zone detection — within a unified real-time processing pipeline.
- Deployed models using Docker on Linux servers, processing real-time video at 30+ FPS with optimized, low-latency inference pipelines.

## PROJECTS

**AI Interview Bot — Real-Time Voice Interview Practice Platform** | *Next.js, FastAPI, Python, Gemini AI, Speech Recognition*
github.com/ashutoshbari/AI-Interview-Bot

- Built to help candidates practice real interviews in a natural, spoken way instead of just reading questions off a screen.
- Designed an automatic voice conversation flow — the AI asks a question via text-to-speech, listens to the spoken answer using speech recognition, and moves to the next question — functioning as a lightweight conversational agent.
- Orchestrated a 7-stage stateful interview workflow powered by the Gemini LLM — dynamically generating context-aware questions, evaluating candidate responses, and adapting the flow in real time, then auto-generating a PDF performance report.

**Real-Time Smoking Detection System** | *YOLOv8, OpenCV, Streamlit*
github.com/ashutoshbari

- Built to automatically flag smoking in restricted areas from live video/RTSP camera feeds instead of relying on manual monitoring.
- Trained a YOLOv8 model to detect smoking activity in real time and trigger instant alerts with a cropped snapshot whenever it was detected.
- Built a Streamlit dashboard so users could monitor live feeds, review alert history, and view basic analytics in one place.

**Battery Health Prediction** | *Scikit-learn, Flask*
github.com/ashutoshbari

- Built to estimate battery health from usage data, helping predict maintenance needs before failure.
- Trained and compared Linear Regression and Random Forest models on battery data, evaluating them using $R^2$ score and MSE to pick the best-performing model.
- Wrapped the model in a Flask REST API with a simple web frontend so users could input data and get real-time health predictions.

## EDUCATION

**Bachelor of Engineering in Electronics & Telecommunication** *2021 – 2025*

Savitribai Phule Pune University | CGPA: 7.2/10 | HSC: 85.60% | SSC: 81.20%